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The asymmetric travelling salesman problem: on generalizations of disaggregated Miller–Tucker–Zemlin constraints

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Abstract

In this paper we show that a multicommodity flow (MCF) model can be aggregated into a node-oriented model which in turn, can be seen as a disaggregation of the well-known Miller–Tucker–Zemlin model. Several outcomes of this node-oriented aggregation are also discussed: (i) the derivation of an "augmented" MCF model with a tighter linear programming (LP) relaxation and which is obtained by adding to MCF a disaggregated version of the Desrochers and Laporte inequalities together with a suitable set of linking constraints and (ii) the derivation of generalizations of the disaggregated Miller–Tucker–Zemlin constraints for paths. These generalized constraints can then be used to show that the LP relaxation of the new and tighter MCF model implies an exponentially sized set of lifted circuit inequalities (simple FD inequalities) which are known to be facet defining for the asymmetric travelling salesman polytope. Generalizations of the disaggregated Desrochers and Laporte inequalities which tighten the LP relaxation of the augmented MCF model are also proposed. © 2001 Elsevier Science B.V. All rights reserved.

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1. Introduction

Given a loop-free directed graph $G = (V,A)$ where $V = \{1,\ldots,n\}$ and costs c_{ij} for each arc $(i,j) \in A$ the asymmetric travelling salesman problem (ATSP) (see [14]) is to

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find the least cost Hamiltonian circuit (an elementary circuit traversing all nodes) contained in the graph. In Section 2, we survey models for the ATSP which are relevant for the development of this paper. Among the surveyed models we include a model involving circuit inequalities and an equivalent (in terms of the corresponding linear programming (LP) relaxations) model based on a disaggregation of the Miller–Tucker–Zemlin constraints. In Section 3, we show that one of these disaggregated models is a node-oriented aggregation of a multicommodity flow (MCF) model and use this result to derive an augmented MCF model with a tighter LP relaxation. In Section 4, we show that the aggregation presented in Section 3 suggests generalizations of the inequalities involved in the disaggregated Miller–Tucker–Zemlin (DMTZ) models. These generalizations are then used to show that the LP relaxation of the tighter MCF model implies an exponentially sized set of lifted circuit inequalities which are facet defining for the asymmetric travelling salesman (ATS) polytope. In Section 5, we also show that inequalities which apparently are new for the ATSP are suggested by suitable combinations of the generalized constraints. In the remainder of this paper, let P_L denote the LP relaxation of formulation P. Additionally, let $F(P)$ denote its set of feasible solutions and let $v(P)$ denote the cost of its optimal solution.

2. (Lifted) circuit inequalities and corresponding extended formulations

Many valid formulations for the ATSP are based on the following general scheme (see [13]):

$$\min \quad \sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}X_{ij} \tag{1}$$

$$\text{s.t} \quad \sum_{i=1}^{n} X_{ij} = 1, \quad j = 1,\ldots,n, \tag{2a}$$

$$\sum_{j=1}^{n} X_{ij} = 1, \quad i = 1,\ldots,n, \tag{2b}$$

$$\{(i,j)\colon i,j = 2,\ldots,n; X_{ij} = 1\} \text{ do not contain subtours}, \tag{3}$$

$$X_{ij} \in \{0,1\}, \quad i,j = 1,\ldots,n, \tag{4}$$

where the binary arc variables X_{ij} indicate whether arc (i,j) is in the optimal tour. For simplicity, all the models presented in this work are defined in a complete graph and we consider $X_{ij} = 0$ for all the arcs (i,j) not in A. Constraints (2a), (2b) and (4) define the usual assignment relaxation for the ATSP. Constraints (3) prevent subtours not containing node 1 and can be written in several different forms. Together with constraints (2a) and (2b) they guarantee that subtours containing the node 1 are also

not allowed. The LP relaxation of each model discussed in this paper is obtained by replacing constraints (4) by

$$0 \leqslant X_{ij} \leqslant 1, \quad i,j = 1,\ldots,n. \tag{4'}$$

Several different ways of modelling constraints (3) are discussed in [16,13] and more recently in [10]. Constraints (3) can be modelled either by using inequalities involving only the X_{ij} variables (such formulations are called "natural" formulations because they contain one and only one variable corresponding to each arc included in the underlying graph) or by using inequalities involving additional variables (such formulations are called "extended formulations"). The information attached to these additional variables may reduce considerably the number of constraints involved in the model and quite often we can derive compact formulations (i.e., formulations involving a polynomial number of constraints and variables). By projecting the set of feasible solutions of the LP relaxation of an extended formulation into the subspace defined by the X_{ij} variables one obtains a natural formulation which is equivalent to the first one in the sense that both formulations produce the same LP bound. However, such natural formulations usually involve an exponential number of constraints.

A recent extended and compact formulation for the ATSP was given in [10]. Consider the binary variables V_{ki} $(i,k = 2,\ldots,n; i \neq k)$ indicating whether node k is in the path from node 1 to node i, and the following system of inequalities:

$$X_{ij} + V_{ki} \leqslant V_{kj} + 1, \quad i,j,k = 2,\ldots,n; \ \ k \neq j \ \text{ and } \ k \neq i, \tag{5a}$$

$$X_{ij} \leqslant V_{ij}, \quad i,j = 2,\ldots,n; \ \ i \neq j, \tag{5b}$$

$$X_{ij} + V_{ji} \leqslant 1, \quad i,j = 2,\ldots,n; \ \ i \neq j, \tag{5c}$$

$$V_{kj} \in \{0,1\}, \quad k,j = 2,\ldots,n; \ \ k \neq j. \tag{6}$$

By adding (5a)–(6) to (2a), (2b) and (4) we obtain a valid model for the ATSP which will be denoted by DMTZ. This designation follows from the fact that system (5a)–(5c) can be seen as a disaggregation of the well-known subtour elimination constraints given in [15] (see [10]). Constraints (5a) state that node k is in the path from nodes 1 to j whenever arc (i,j) is included in the optimal solution and node k is in the path from node 1 to node i. Constraints (5b) state that node i is in the path from node 1 to node j if arc (i,j) is in the optimal solution. Finally, constraints (5c) state that arc (i,j) is in the optimal solution if and only if node j is not in the path from nodes 1 to i. The binary conditions (6) were introduced in order to give an intuitive and precise meaning to the system defined by (5a)–(5c). However, as shown in [10], (6) can be replaced by

$$V_{kj} \in \mathbb{R}, \quad k,j = 2,\ldots,n; \ \ k \neq j \tag{6'}$$

in the LP relaxation of the model without altering its optimal value. Gouveia and Pires [10] have also shown that the projection into the X_{ij} subspace of the set $F(\mathrm{DMTZ_L})$ is

completely described by (2a), (2b), (4′) and the following set of "circuit elimination" constraints

$$\sum_{(i,j)\in A} X_{ij} \leqslant |C| - 1 \quad \forall C \text{ in } G_{V_1}, \tag{7}$$

where G_{V_1} is the complete graph induced by the set of nodes $V_1 = \{2,\ldots,n\}$ and C is any set of arcs defining an elementary circuit (i.e., no nodes are repeated in the circuit) in G_{V_1}. These constraints have been described in [12] and state that the maximum number of arcs which can be packed in a given circuit C is given by $|C|-1$. The formulation given by (1), (2a), (2b), (7) and (4) will be denoted by CIRCUIT and is an example of an exponentially sized natural formulation whose LP relaxation results from projecting the LP relaxation of an extended and compact formulation. We say that $F(\mathrm{DMTZ_L})$ is an extended characterization of $F(\mathrm{CIRCUIT_L})$. The circuit constraints (7) can be lifted into the well-known "subtour elimination" constraints

$$\sum_{i\in S}\sum_{j\in S} X_{ij} \leqslant |S| - 1 \quad \forall |S| \subseteq \{2,\ldots,n\} \text{ and } |S| \geqslant 2 \tag{8}$$

which are known to be facet defining for the ATS polytope whenever $|S| \leqslant n-2$ and $n \geqslant 5$ (see [12]). Notice that any constraint (8) with $|S| = n-1$ is implied by the assignment constraints (2a) and (2b). Constraints (8) can also be seen as clique packing inequalities because they state that the maximum number of arcs which can be packed in the clique defined by the set of nodes S is given by $|S|-1$. Therefore, we shall use the term "clique" constraints when refering to constraints (8) and model (1), (2a), (2b), (8) and (4) will be denoted by CLIQUE.

The interest in examining the weaker set of circuit constraints (7) is that these inequalities can also be lifted in other different ways. In fact, several new classes of facet-defining inequalities for the ATS polytope can be generated by applying sequential lifting (see [12]) to the circuit inequalities (7). Briefly, the sequential lifting procedure works as follows. All the coefficients for the arcs not in the circuit are initially set to $-\infty$. Then, they are sequentially increased, one at a time, up to a maximum value for which the inequality remains valid for the ATS polytope. It is known that the coefficient of an arc (i,j) such that i or j are not in the circuit is equal to 0. Furthermore, the first lifted arc (i,j) such that (i,j) is not in the circuit C but both i and j are in the circuit node set, receives a coefficient of 2. The remaining arcs have coefficients in the set $\{0,1,2\}$. Different sequences for lifting the arcs may lead to different lifted circuit inequalities. The value of a given coefficient is largest if lifted first. As a simple example of a sequentially lifted circuit inequality, consider the circuit inequality (7) associated to the circuit $\{(p,q),(q,r),(r,p)\}$ of cardinality three. If the arc (q,p) is lifted first it receives a coefficient of value two and the remaining arcs receive a coefficient of value 0. The lifted inequality is

$$(2X_{qp}+)X_{pq} + X_{qr} + X_{rp} \leqslant 2 \tag{9}$$

with the lifted part being shown in parentheses. Notice that two other inequalities can be lifted from the same circuit inequality if we consider either arc (r,q) or arc (p,r)

to be lifted first. However, up to an isomorphism the three inequalities are identical. The inequality above is known to be facet defining for the ATS polytope (see [12,7]) if $n \geqslant 4$. Several families of lifted circuit inequalities which are facet-defining for the ATS polytope are described in [12] and more recently in [2].

As pointed out before, $F(\mathrm{DMTZ_L})$ gives an extended characterization of $F(\mathrm{CIRCUIT_L})$ (in fact, the proof in [10] can be slightly changed to show that the set (5a)–(5c) and (6′) gives an extended characterization of (7)). It would be interesting to check whether some of the lifted circuit inequalities mentioned before can be obtained by projecting in the X_{ij} space liftings of the inequalities in set (5a)–(5c). We start by pointing out that constraints (5b) and (5c) cannot be lifted. To see this note that by summing up constraint (5b) for a given pair (i,j) with constraint (5c) for the pair (j,i) we obtain the facet-defining clique inequality (8) for the set $S=\{i,j\}$ (in this case, equivalent to the circuit inequality (7) for the circuit $\{(i,j),(j,i)\}$). If constraint (5b) and/or constraint (5c) could be lifted, then the same argument would yield an inequality which is stronger than a facet-defining inequality. Thus, (5b) and (5c) cannot be lifted. On the other hand, two different ways of lifting constraints (5a) have been proposed in [10]

$$(X_{ji}+)X_{ij} + V_{ki} \leqslant V_{kj} + 1, \quad i,j,k = 2,\ldots,n; \;\; k \neq i,j, \tag{10}$$

$$(X_{kj} + X_{ik}+)X_{ij} + V_{ki} \leqslant V_{kj} + 1, \quad i,j,k = 2,\ldots,n; \;\; k \neq i,j. \tag{11}$$

Again, we have inserted the lifted terms in parentheses. Constraints (10) and (11) are disaggregations of inequalities proposed, respectively, by Desrochers an Laporte [5] and [9] which are also liftings of the original Miller–Tucker–Zemlin constraints. The validity of (10) follows from the fact that $X_{ji} + X_{ij} \leqslant 1$ is always feasible and that $V_{ki} = 1$ iff $V_{kj} = 1$ whenever $X_{ji} + X_{ij} = 1$. With respect to (11) notice that if $X_{ij} = 1$ then $X_{kj} = X_{ik} = 0$ and the validity of (11) follows from the validity of (5a). When $X_{ij} = 0$, (11) is still valid because it results from adding (5b) for the pair (k,j) to (5c) for the pair (i,k). In the following, let the model DMTZ with (5a) replaced by (10) be denoted by D1MTZ and let the model DMTZ with (5a) replaced by (11) be denoted by D2MTZ.

The projection of $F(\mathrm{D1MTZ_L})$ and $F(\mathrm{D2MTZ_L})$ into the space of the X_{ij} variables produces liftings of the circuit inequalities (7). However, such liftings are not interesting because they correspond to inequalities which are dominated by the clique constraints (8). In fact the characterizations of $F(\mathrm{D1MTZ_L})$ and $F(\mathrm{D2MTZ_L})$ into the space of the X_{ij} variables (see [10]) simply show that the LP relaxation of these two models is dominated by the LP relaxation of the CLIQUE model. It is interesting to point out that $F(\mathrm{D2MTZ_L})$ gives an extended description of $F(\mathrm{CLIQUE_L})$ when $n = 5$. To see this notice that the clique constraint on the set $S = \{2,3,4,5\}$ is implied by the assignment constraints (2a) and (2b). Clique constraints on sets of cardinality two contained in $\{2,3,4,5\}$ are implied by constraints (5b) and (5c). Finally, by adding constraint (11) for (i,j,k) to constraint (11) for (j,i,k) we obtain a clique constraint on the set $\{i,j,k\}$. As this is valid for all $i,j,k = 2,\ldots,5$ we obtain all the clique

constraints of cardinality 3. Notice also that the weaker $F(\mathrm{DMTZ_L})$ gives an extended description of $F(\mathrm{CLIQUE_L})$ when $n = 4$.

A more interesting model, D3MTZ, has also been proposed in [10] and is obtained by using together the two sets of inequalities (10) and (11). One can simply see that one of the lifted inequalities (9) is obtained by adding constraint (10) for the triple (i,j,k) with constraint (11) for the triple (j,i,k). The same reasoning applied to all triples (i,j,k) $(i,j,k = 2,\ldots,n)$ shows that the LP relaxation of this model satisfies all the inequalities (9) for triples (i,j,k) $(i,j,k = 2,\ldots n)$. Then, the model $\mathrm{CLIQUE_L}$ does not dominate the model $\mathrm{D3MTZ_L}$ because the $\mathrm{D3MTZ_L}$ model satisfies facet-defining inequalities for the ATS polytope which are not satisfied by $\mathrm{CLIQUE_L}$. These arguments together with what has been stated in the previous paragraph show that for $n = 5$, we have $v(\mathrm{D3MTZ_L}) \geqslant v(\mathrm{CLIQUE_L})$. Furthermore, strict inequality holds for some instances (an example of such an instance is given in the next section).

The previous discussion suggests that other interesting, or even new, inequalities for the ATSP involving only the X_{ij} variables may possibly be generated by combining adequately generalizations of the lifted inequalities (10) and (11). In fact, we will show later that an exponentially sized set of lifted k-circuit inequalities for $k \geqslant 3$ ("lifted k-circuit inequality" is used to denote an inequality which is lifted from a circuit inequality (7) associated to a circuit of cardinality k) can be obtained by combining constraints (10) with generalizations of constraints (11). It will also be shown that these generalized constraints can be generated from a MCF model. Generalizations of constraints (10) will also be suggested and inequalities which seem to be new will be generated from these generalized inequalities.

3. Aggregating the LP relaxation of a multicommodity flow model into D2MTZL

A well-known example of an extended and compact formulation for the ATSP whose LP relaxation gives an extended and compact representation of the set $F(\mathrm{CLIQUE_L})$ is discussed in this section. Furthermore, we shall present a relationship between its LP relaxation and the LP relaxation of some of the models discussed in the previous section. Consider the additional set of flow variables F_{kij} which indicate whether arc (i,j) is in the path from node k $(k = 2,\ldots,n)$ to node 1. Consider, also, the following set of flow balancing constraints together with an adequate set of $O(n^3)$ constraints which relate the new flow variables with the arc design variables:

$$\sum_{i=1}^{n} F_{kji} - \sum_{i=1}^{n} F_{kij} = 0, \quad j,k = 2,\ldots,n; \;\; j \neq k, \tag{12a}$$

$$\sum_{i=1}^{n} F_{jji} = 0, \quad j = 2,\ldots,n, \tag{12b}$$

$$F_{kij} \leqslant X_{ij}, \quad i,j = 1,\ldots,n; \;\; k = 2,\ldots,n, \tag{13}$$

$$F_{kij} \geqslant 0, \quad i,j = 1,\ldots,n; \;\; k = 2,\ldots,n. \tag{14}$$

Let MCF denote the model defined by (2a), (2b), (12a), (12b), (13), (4) and (14). To simplify the indexing, the variables F_{jij} $(i = 1,\ldots,n;\ j = 2,\ldots,n)$ and F_{k1j} $(j,k = 2,\ldots,n)$ are not included in the model (clearly, arc (i,j) is not used in the path from j to node 1 and no arc $(1,j)$ is used in any path to node 1). The $n-1$ flow balancing constraints for node 1 are omitted from the system given above because they are redundant in the corresponding LP relaxation. The model MCF has been proposed by Claus [4]. There are many ways for deriving a MCF model for the TSP (see [13]). As far as we know, Wong [16] was the first to propose a MCF model for the ATSP and, more importantly, was the first to establish an equivalence relationship between the LP relaxation of a MCF model with the LP relaxation of the natural formulation CLIQUE. In fact, a simple application of the Max Flow–Min Cut theorem (see [1]) together with some constraint manipulation using either constraints (2a) or (2b) shows that the projection into the X_{ij} subspace of the set of feasible solutions defined by the LP relaxation of the MCF model is completely described by CLIQUE_L.

This equivalence result and the relationships between the model CLIQUE_L and three models D1MTZ_L, D2MTZ_L and D3MTZ_L mentioned in the previous section show that MCF_L dominates each of the two models D1MTZ_L and D2MTZ_L, but does not dominate D3MTZ_L. The dominance between MCF_L and each of the two models D1MTZ_L and D2MTZ_L implies that for a feasible solution $\{X_{ij}, F_{kij}\}$ for MCF_L, there exists an assignment of values $V1_{ki}$ and $V2_{ki}$ such that $\{X_{ij}, V1_{ki}\}$ is feasible for D1MTZ_L and $\{X_{ij}, V2_{ki}\}$ is feasible for D2MTZ_L. On the other hand, the non-dominance between MCF_L and D3MTZ_L implies that the $V1_{ki}$ values may be different from the $V2_{ki}$ values for at least one pair (i,k). Solving D3MTZ_L is the same as solving the system defined by D1MTZ_L rewritten with the $V1_{ki}$ variables together with the system defined by D2MTZ_L rewritten with the $V2_{ki}$ variables plus an additional set of equalities $V1_{ki} = V2_{ki}$, one for each pair (k,i) (clearly, inequalities (5b) and (5c) are duplicated and one of the copies may be dropped). This additional set of inequalities tightens the combined model in such a way that it is no longer dominated by MCF_L. As an illustration, consider the following solution taken from Gouveia and Pires (1999a):

Solution: $X_{12} = X_{14} = X_{23} = X_{24} = X_{31} = X_{32} = X_{41} = X_{43} = \frac{1}{2}$;

other variables equal to zero.

The above solution is feasible for MCF_L. It satisfies (2a), (2b), (4′) and the subtour elimination constraints (8). Therefore, there exists an assignment of values $V1_{ki}$ and $V2_{kj}$ such that $\{X_{ij}, V1_{ki}\}$ is feasible for D1MTZ_L and $\{X_{ij}, V2_{ki}\}$ is feasible for D2MTZ_L. However, it is not difficult to see that $V1_{42}$ should be equal to $\frac{1}{2}$ and $V2_{42}$ should be equal to 0. This simply means that the above solution in the X_{ij} variables cannot be feasible for D3MTZ_L.

The above arguments suggest three hypothetical models whose LP relaxations are tighter than the LP relaxation of MCF. Assume that we know a linear transformation

LT1(F_{kij}) involving the flow variables F_{kij} and such that

$$V1_{ki} = \mathrm{LT1}(F_{kij}) \quad \text{for each } (k,i). \tag{15}$$

The above arguments show that the value of the $V1_{ki}$ variables given by (15) may not be feasible for (11) rewritten with the $V1_{ki}$ variables. Thus, solutions which are feasible in $\mathrm{MCF_L}$ solution may become unfeasible when (15) and (11) are added to MCF. This gives our first hypothetical model which is obtained by adding (15) and (11) to MCF.

The second hypothetical model is based on the assumption that we can produce a different linear transformation LT2(F_{kij}) such that

$$V2_{ki} = \mathrm{LT2}(F_{kij}). \tag{16}$$

By a similar reasoning we can show that the optimal $\mathrm{MCF_L}$ solution can be cut away by adding to $\mathrm{MCF_L}$, (16) and the first set of lifted inequalities (10) rewritten with the variables $V2_{ki}$.

The two linear transformations can be used to obtain a third hypothetical model which subsumes the other two. This model is simply obtained by adding to $\mathrm{MCF_L}$ the two transformations (15) and (16) plus the equality constraints $V1_{ki}=V2_{ki}$, one for each pair (k,i). Clearly, these three proposed models are hypothetical in the sense that in order to derive such models we have to produce the two involved linear transformations LT1(F_{kij}) and LT2(F_{kij}).

We have been unable to find the linear transformation LT1(F_{kij}). Furthermore, it is not clear if such a transformation does necessarily exist. Therefore, two of the previous models will remain hypothetical for the time being. On the other hand, the other linear transformation LT2(F_{kij}) is quite straightforward. Consider, the following linear transformation:

$$\sum_{j=1}^{n} F_{kij} = V_{ki}, \quad i,k = 2,\ldots,n. \tag{17}$$

Notice that the left-hand side of (17) counts the number of arcs diverging from node i in the path from node k to node 1. Under the other constraints involved in the MCF model this number is either equal to 1 (i.e., node k is in the path to node i and $V_{ki}=1$) or is equal to 0 (and $V_{ki}=0$). We shall show next that $\mathrm{D2MTZ_L}$ can be derived from $\mathrm{MCF_L}$ under (17). More precisely we shall show that the projection of $F(\mathrm{MCF_L})$ into the (X_{ij},V_{ki}) space under the linear transformation (17) is strictly contained in $F(\mathrm{D2MTZ_L})$. First, we show how to derive constraints (5b) and (5c) from $\mathrm{MCF_L}$. With respect to (5b), notice that in the presence of constraints (2b) and (12b) it is easy to prove that $F_{jji}=X_{ji}$ for each arc (j,i) (if $F_{jji}<X_{ji}$ for a given pair (j,i) then we shall have $F_{jjp}>X_{jp}$ for a different pair (j,p) contradicting (13)). Then by using (17) and weakening (12a) for j,k we obtain

$$V_{kj} = \sum_{q=1}^{n} F_{kjq} = \sum_{q=1}^{n} F_{kqj} \geqslant F_{kkj} = X_{kj}, \quad j,k = 2,\ldots,n$$

which gives (5b). To show that (5c) can be derived in a similar manner, we add (13) for $j = 1,\ldots,n$ ($j \neq k$). Then, by noticing that variable F_{kij} is not defined for $j = k$ and using (17) we obtain

$$V_{ki} = \sum_{q=1;q\neq k}^{n} F_{kiq} \leqslant \sum_{q=1;q\neq k}^{n} X_{iq} = 1 - X_{ik}, \quad i,k = 2,\ldots,n$$

which is precisely (5c). With respect to (11) we start by weakening (12a) into

$$\sum_{q=1}^{n} F_{kjq} \geqslant F_{kij} + F_{kkj}, \quad i,j,k = 2,\ldots,n;\ \ j \neq k;\ \ i \neq k \tag{12aw}$$

which is equivalent to

$$V_{kj} \geqslant F_{kij} + X_{kj}, \quad i,j,k = 2,\ldots,n;\ \ j \neq k;\ \ i \neq k \tag{18}$$

under (17) (notice again that $F_{kkj} = X_{kj}$). By adding

$$\sum_{q=1;q\neq j}^{n} F_{kiq}$$

to each member of (18), grouping adequate terms and using (17) we obtain

$$\sum_{q=1;q\neq j}^{n} F_{kiq} + V_{kj} \geqslant V_{ki} + X_{kj}, \quad i,j,k = 2,\ldots,n;\ \ j \neq k;\ \ i \neq k. \tag{19}$$

By using (13) and using the fact that F_{kiq} is not defined for $q = k$ we obtain

$$\sum_{q=1;q\neq j;q\neq k}^{n} X_{iq} + V_{kj} \geqslant V_{ki} + X_{kj}, \quad i,j,k = 2,\ldots,n;\ \ j \neq k;\ \ i \neq k. \tag{20}$$

Finally, by using (2b) we obtain

$$X_{ij} + (X_{ik} + X_{kj}) + V_{ki} \leqslant 1 + V_{kj}, \quad i,j,k = 2,\ldots,n;\ \ j \neq k;\ \ i \neq k \tag{21}$$

which is equivalent to (11). A derivation of weaker (5a) constraint could have been obtained if the term "$+F_{kkj}$" is dropped from (12aw) and if we had not used the fact that F_{kiq} is not defined for $q = k$ when tranforming (19) into (20).

The derivation of (11) from $\mathrm{MCF_L}$ under the linear transformation (17) shows that the same transformation cannot be used to generate the disaggregated Desrochers and Laporte inequalities (10) from $\mathrm{MCF_L}$. Therefore, (10) and (17) are not redundant in $\mathrm{MCF_L}$ and a new model with a tighter LP relaxation can be obtained by adding (10) and (17) to MCF (the new model will be denoted by MCF+ in the sequel). One interesting point of research is to find a description of the LP relaxation of MCF+ into the space of the natural variables X_{ij}. Note that the clique constraints (8) are contained in such a description because they are implied by the $\mathrm{MCF_L}$ model. The same happens with the lifted circuit inequalities (9) for all $i,j,k = 2,\ldots,n$. This follows from the fact that these inequalities result from adding adequately constraints (10) to constraints (11) and that constraints (11) are implied by $\mathrm{MCF_L}$ under (17). We shall show in the next section that $\mathrm{MCF+_L}$ implies an exponentially sized set of lifted k-circuit inequalities for all k ($3 \leqslant k \leqslant n-1$).

4. Generalizing inequalities (11)

We note that generalizations of constraints (11) which might tighten the LP relaxation of the models DMTZ and D2MTZ might be found by carefully examining the derivation of $D2MTZ_L$ from the MCF_L model. Notice that each constraint (12aw) for i,j,k is substantially weaker than the corresponding constraint (12a) for j,k. Additionally, we have generated $D2MTZ_L$ from the weaker constraints (12aw) suggesting that we may obtain stronger constraints if we add extra terms to the right-hand side of (12aw) and/or consider at the same time several constraints (12aw). In fact, we shall derive in this way one generalization of constraints (11).

Consider the two following inequalities:

$$\sum_{q=1}^{n} F_{kjq} \geqslant F_{kij} + F_{kmj} + F_{kkj}, \quad i,j,m,k = 2,\ldots,n;\ \ j \neq k;\ \ i \neq k;\ \ m \neq k,$$

$$\sum_{q=1}^{n} F_{kmq} \geqslant F_{kim} + F_{kkm}, \quad i,m,k = 2,\ldots,n;\ \ i \neq k;\ \ m \neq k.$$

By adding these two inequalities and using (17) we obtain

$$V_{kj} + V_{km} \geqslant F_{kij} + F_{kmj} + F_{kim} + X_{kj} + X_{km}.$$

By adding

$$\sum_{q=1;q\neq j;q\neq m}^{n} F_{kiq} + \sum_{q=1;q\neq j}^{n} F_{kmq}$$

to each member of the above inequality, grouping adequate terms and using (17) we obtain

$$\sum_{q=1;q\neq j;q\neq m}^{n} F_{kiq} + \sum_{q=1;q\neq j}^{n} F_{kmq} + V_{kj} \geqslant V_{ki} + X_{kj} + X_{km}.$$

By using (13) and the fact that F_{kiq} and F_{kmq} are not defined for $q = k$ we obtain

$$\sum_{q=1;q\neq j;q\neq m;q\neq k}^{n} X_{iq} + \sum_{q=1;q\neq j;q\neq k}^{n} X_{km} + V_{kj} \geqslant V_{ki} + X_{kj} + X_{km}.$$

Finally, by using (2b) we obtain

$$1 - X_{ij} - X_{ik} - X_{im} + 1 - X_{mj} - X_{mk} + V_{kj} \geqslant V_{ki} + X_{kj} + X_{km}$$

which can be rewritten as

$$X_{im} + X_{mj} + X_{ij} + X_{ik} + X_{km} + X_{mk} + X_{kj} + V_{ki} \leqslant V_{kj} + 2$$

$$i,m,j,k = 2,\ldots,n;\ \ k \neq i,m,j. \tag{22}$$

Notice that (22) for a quadruple (i,m,j,k) is obtained by adding the two lifted constraints (11) for the triples (i,m,k) and (m,j,k) and subsequently, inserting the

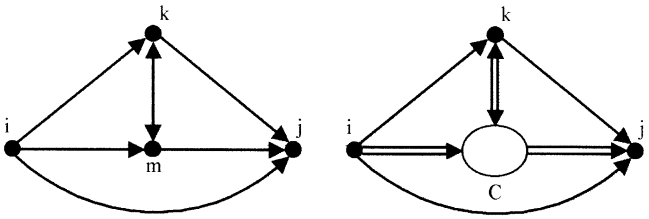


Fig. 1. (a) Support graph of the 2-path inequalities (22) and (b) their generalization (23). The arcs in the clique C are not shown.

term "$+X_{ij}$" in the corresponding left-hand side. Fig. 1a depicts the support graph associated to the X variables on the left-hand side of (22).

The argument used above to derive the 2-path inequalities (22) from $\mathrm{MCF_L}$ can be used to generalize these inequalities to 2-paths where the middle node, node m, is replaced by a clique (see Fig. 1b). The generalized constraints are written as

$$V_{ki} + \sum_{m\in C}(X_{im} + X_{km}) + \sum_{m\in C}(X_{mj} + X_{mk})$$

$$+ \sum_{p;q\in C} X_{pq} + X_{ik} + X_{kj} + X_{ij} \leqslant 1 + |C| + V_{kj}$$

$$i,j,k = 2,\ldots,n;\ \ i,j \neq k;\ \ C \subset V \setminus \{1,i,j,k\}. \tag{23}$$

Notice that (23) contains (11) ($|C| = 0$) and (22) ($|C| = 1$) as special cases. These constraints can also be derived from $\mathrm{MCF_L}$ and thus, they are valid. For simplicity, we only sketch their derivation from $\mathrm{MCF_L}$. Consider the following two inequalities. The first inequality is a weakened flow conservation constraint (12a) for node j and the second constraint results from adding adequate weakened constraints (12a) for all nodes $m \in C$

$$\sum_{q=1}^{n} F_{kjq} \geqslant \sum_{m\in C} F_{kmj} + F_{kij} + F_{kkj}, \quad i,j,k = 2,\ldots,n;\ \ i,j \neq k;\ \ C \subset V \setminus \{1,i,j,k\}$$

$$\sum_{m\in C}\sum_{q=1}^{n} F_{kmq} \geqslant \sum_{m\in C}\left(\sum_{n\in C} F_{knm} + F_{kim} + F_{kkm}\right)$$

$$i,k = 2,\ldots,n;\ \ i \neq k;\ \ C \subset V \setminus \{1,i,j,k\}.$$

By adding the two above inequalities, using (17) and the fact $F_{kkp}{=}X_{kp}(\,p \in C\cup\{j\})$ we obtain

$$V_{kj} + \sum_{m\in C} V_{km} \geqslant F_{kij} + X_{kj} + \sum_{m\in C}\left(F_{kmj} + \sum_{n\in C} F_{knm} + F_{kim} + X_{km}\right)$$

$$i,j,k = 2,\ldots,n;\ \ i \neq k;\ \ C \subset V \setminus \{1,i,j,k\}.$$

By adding

$$\sum_{q=1;q\neq j;q\notin C}^{n} F_{kiq} + \sum_{m\in C}\sum_{q=1;q\neq j;q\notin C}^{n} F_{kmq}$$

to each member of the above inequality, cancelling equal terms, using the fact that F_{kmq} $(m \in C)$ are not defined for $q = k$ and rearranging we obtain (23).

If we add the disaggregated Desrochers and Laporte inequality (10) for a triple (j,i,k) to inequality (23) for the 4-tuple (i,j,k,C) we obtain the following inequality:

$$\sum_{m\in C}(X_{im}+X_{km}) + \sum_{m\in C}(X_{mj}+X_{mk}) + \sum_{p;q\in C} X_{pq} + X_{ik} + X_{kj} + 2X_{ij} + X_{ji} \leqslant 2 + |C|$$

$$i,j,k = 2,\ldots,n;\ \ i,j \neq k;\ \ C \subset V \setminus \{1,i,j,k\} \tag{24}$$

which can be seen as lifted from the $(|C|+3)$-circuit inequality corresponding to the circuit $\{(k,j),(j,i),(i,s_1),(s_1,s_2),\ldots,(s_{|C|-1},s_{|C|}),(s_{|C|},k)\}$ where $C=\{s_1,\ldots,s_{|C|}\}$. Again, we point out that different lifted inequalities which are isomorphic to (24) can also be obtained by considering different sequences for the lifted coefficients. Notice also that in general, the same inequality can be lifted from different inequalities corresponding to different circuits. For instance, (24) can be seen as lifted from the circuit inequality where the nodes in C are ordered in a different way. Inequality (24) can also be seen as lifted from the inequality associated to the circuit $\{(k,s_1),(s_1,s_2),\ldots,(s_{|C|-1},s_{|C|}),(s_{|C|},j),(j,i),(i,k)\}$.

The lifted inequalities (24) are special cases of the simple FD inequalities introduced by Balas and Fischetti [3] and shown to be facet defining for the ATS polytope in the same work. In order to define such constraints, Balas and Fischetti [3] require a partition of the node set V into four subsets $S,H,T1$ and $T2$ such that node $1 \in S$. Then, by making $H=\{i\}, T1=C\cup\{k\}, T2=\{j\}$ and S equal to the set defined by the remaining nodes, it is easily checked that inequalities (24) are the simple facet-defining inequalities when $|H|=|T2|=1$.

As constraints (23) are generated from the model $\mathrm{MCF_L}$ and constraints (10) are included in MCF+, we can conclude that the LP relaxation of MCF+ also implies all of the lifted circuit inequalities (24). Thus, MCF+ is a compact model which subsumes this particular case (but nonetheless, exponential sized set) of simple FD inequalities. Notice that inequality (9) corresponds to (24) with $|C|=0$. The case $|C|=1$ has already been discussed in the literature (see [12,2]).

In the following, the model D2MTZ with all constraints (23) will be denoted by GD2MTZ. In theory, we should be able to generate a set of new inequalities (NI) defined on the variables X_{ij} and V_{ki} such that $\{(X_{ij},V_{ki})$: (X_{ij},V_{ki}) satisfy (2a), (2b), (4′), (6′) and (NI)$\}$ is the projection of $F(\mathrm{MCF_L})$ into the (X_{ij},V_{ki}) space under the linear transformation (17). Then, the projection into the natural X_{ij} space of the first polytope is given precisely by $F(\mathrm{CLIQUE_L})$. While, we believe that (5b), (5c) and (23) are included in that description, these inequalities are not sufficient for defining the set (NI). In fact, it is not difficult to find ATSP instances with $n \geqslant 5$ which are feasible for $\mathrm{G2DMTZ_L}$ but are not feasible for $\mathrm{MCF_L}$ (we have already pointed out

that $F(\mathrm{G2DMTZ_L})$ gives an extended description of $F(\mathrm{CLIQUE_L})$ when $n = 5$. However, this does not necessarily mean that $F(\mathrm{G2DMTZ_L})$ describes the projection of $F(\mathrm{MCF_L})$ into (X_{ij}, V_{ki}) space under the linear transformation (17). In fact, the previously mentioned instances with five nodes confirm that the set of lifted inequalities (11) do not define (NI) when $n = 5$).

The interest of such a model is that it provides an alternative and possible faster way of computing $v(\mathrm{MCF_L})$. We start by solving the LP relaxation of D2MTZ model of generate inequalities from the new set (NI) as needed until we obtain $v(\mathrm{MCF_L})$. This cutting plane method might prove to be more efficient than solving directly the LP relaxation of the MCF model which takes large amounts of CPU time (see, for instance, [10]). Even an incomplete description of the model may be useful because the inequalities in the incomplete model (such as inequalities (23)) can be used to tighten the lower bounds given by the LP relaxation of the D2MTZ model. While, one might argue that such a cutting plane method would hardly be competitive with available cutting plane methods for the ATSP which involve only natural models (see, for instance, [8]) we believe that the new inequalities may be useful for problems where the extra variables cannot be removed due to extra cost parameters.

One example is the ATSP with link costs and precedence costs which can be seen as a relaxation of the ATSP with precedence constraints. Such constraints can be imposed by setting $V_{ij} = 1$ and/or $V_{ij} = 0$ for some pairs (i, j). Adding a multiplier to each constraint and adding a penalty term to the objective function yields an ATSP with link and precedence costs. We have not seen any other application for this generalization of the ATSP and therefore, we did not look further into it. A related and possibly more interesting is the ring network design described in Gendreau, Labbé and Laporte (1995) [17]. In this problem, a tour (non-necessarily Hamiltonian) connecting several nodes has to be found. Besides the usual link costs, the cost of the network also involves revenue profits incurred by any pair of nodes which are included in the network. Notice that if two nodes i and j are in the network then either i is in the path to j ($V_{ij} = 1$) or j is in the path to i ($V_{ji} = 1$). This indicates that by associating the revenue profits to the V_{ij} variables we can use the models described in this paper to formulate the ring network design problem as a linear integer program (clearly, as the ring may not be Hamiltonian some constraints have to modified and new ones have to be included). A similar problem has been studied in [11] and a comparison between DMTZ-like models and MCF-like models is reported in the same work. For the instances where the revenues dominate the link costs, the LP bounds given by the two classes of models are nearly equivalent and close to the optimum. For such instances, the DMTZ models seem to be preferable because for most of the cases they produce the optimal integer solution in less time than the MCF-like models produce the optimal LP solution. For the instances where the revenues have the same order of magnitude as the link costs, the LP bounds given by the MCF models are still quite good. However, the LP bounds given by the DMTZ models are substantially worse. It is precisely for these cases that it would be interesting to find the set of inequalities (NI). As we pointed out before, the path inequalities (23) are included in (NI) and for

these instances, reasonable improvements on the LP values of the DMTZ models are obtained by including some of inequalities (23). However, the new lower bounds are still far from the lower bounds given by MCF. These results indicate that it might be interesting to find other inequalities in (NI).

Clearly, what has been said above concerning the interest of obtaining the projection of $F(\mathrm{MCF_L})$ into the (X_{ij}, V_{ki}) space under the linear transformation (17), also applies to the stronger model $\mathrm{MCF}+_{\mathrm{L}}$.

5. Generalizing inequalities (10) for cliques

As pointed out before, any given generalized constraint (22) can be seen as a lifting of an inequality which results from adding two inequalities (11). A similar idea can be used to generalize the other set of lifted constraints, i.e., constraints (10). Consider the two following inequalities (10), one for the triple (i,m,k) and the other for the triple (m,j,k). If we add the two inequalities, we obtain the following inequality which is redundant in any model containing the two original constraints (10)

$$V_{ki} + X_{im} + X_{mi} + X_{mj} + X_{jm} \leqslant 2 + V_{kj}.$$

However, this inequality can be strengthened by adding the term "$+X_{ij} + X_{ji}$" to the left-hand side of the above inequality. The strengthened inequality is depicted as

$$V_{ki} + X_{im} + X_{mi} + X_{mj} + X_{jm} + X_{ij} + X_{ji} \leqslant 2 + V_{kj}$$

$$i,m,j,k = 2,\ldots,n;\quad k \neq i,m,j. \tag{25}$$

The validity of (25) follows from the fact that at most two X_{pq} variables can be equal to 1 on the left-hand side of (25). Moreover, the arcs corresponding to those two variables form a path with two arcs. Then $V_{ki} = 1$ iff $V_{kj} = 1$ because node k is not included in the path and nodes i and j are. Notice that the arcs associated to the variables in the left-hand side of (25) form a clique on the set of nodes $\{i,j,m\}$. This suggests a simple way of generalizing these constraints for larger cliques:

$$V_{ki} + \sum_{p\in S}\sum_{q\in S} X_{pq} \leqslant |S| - 1 + V_{kj} \quad i,j,k = 2,\ldots,n;$$

$$S \subseteq \{2,\ldots,n\};\quad |S| \geqslant 2;\quad k \notin S;\quad i,j \in S. \tag{26}$$

Notice that if the value of the summation term is less than $|S| - 1$ than (26) is trivially true. When the summation term is equal to $|S| - 1$ then the arcs associated to the variables which are equal to 1 form a path with $|S| - 1$ arcs and again we have $V_{ki} = 1$ iff $V_{kj} = 1$ because node k is not included in S and nodes i and j are. This shows that (26) is valid. When $|S| = 2$ we obtain the original disaggregated Desrochers and Laporte inequalities (10) and when $|S| = 3$ we obtain inequalities (25). In the following we denote by GD1MTZ the model D1MTZ with all constraints (26). Notice that if we add constraint (26) for a given i,j,k and S to constraint (26) for j,i,k and

S we obtain the clique constraint for the set S. This simply shows that the projection into the X_{ij} space of $F(\mathrm{GD1MTZ_L})$ is contained in $F(\mathrm{CLIQUE_L})$ (we conjecture that the two models are equivalent for the ATSP, i.e, $F(\mathrm{GD1MTZ_L})$ gives an extended characterization of $F(\mathrm{CLIQUE_L})$).

With respect to the ATSP, and assuming that our conjecture is valid, the new model GD1MTZ does not seem to be interesting because it involves the same number of constraints as CLIQUE but contains more variables. The interest of the GD1MTZ model may arise for problems where the V_{ij} variables cannot be removed as the ATSP with precedence costs or the ring network design problem mentioned before. Notice, however, that $F(\mathrm{GD1MTZ_L})$ cannot be the model which results from projecting $F(\mathrm{MCF_L})$ into the (X_{ij}, V_{ki}) space under the linear transformation (17). This simply follows from the fact that constraints (26) for $|S| = 2$ (the original lifted constraints (10) which are included in MCF+) cannot be derived from $\mathrm{MCF_L}$ under (17).

Consider now, the model MCF++ which is obtained by adding to MCF, constraints (17) and constraints (26). Clearly, the LP relaxation of MCF++ dominates the LP relaxation of MCF+ (because this model only contains constraints (26) for $|S| = 2$). We have already shown that by adequately combining (26) for $|S| = 2$ (i.e. constraint (10)) with constraints (23) we obtain lifted circuit inequalities which are facet-defining for the ATSP. It might be interesting to see what can be obtained by combining inequalities (26) for larger sets S with some of constraints (23). As one simple example, consider the inequality which is obtained by adding constraint (23) for $|C| = 0$ and i, j, k (i.e., constraint (11) for i, j, k) with constraint (26) for j, i, k and $|S| = \{i, j, m\}$:

$$X_{kj} + X_{ik} + X_{im} + X_{mi} + X_{mj} + X_{jm} + 2X_{ij} + X_{ji} \leqslant 3.$$

Fischetti [6] has pointed out that the above inequality is not facet defining but that it results from adding two facet-defining inequalities and subsequently dividing by 3

$$X_{jm} + X_{mi} + X_{ik} + X_{kj} + 2X_{ij} + 2X_{ji} \leqslant 3,$$

$$2X_{jm} + 2X_{mi} + 2X_{ik} + 2X_{kj} + 4X_{ij} + X_{ji} + 3X_{mj} + 3X_{im} \leqslant 6.$$

The first inequality is known from the literature (see Fig. 8.7 (b4) in [12]) and is also sequentially lifted from the circuit inequality associated to the 4-circuit $\{(j, m), (m, i), (i, k), (k, j)\}$. The last inequality can also be seen as a lifting of the same circuit inequality (this is easily seen if one divides it by 2) and according to Fischetti [6] it is facet defining. However, it cannot be generated from the circuit inequality by sequential lifting (notice the coefficient with value $\frac{1}{2}$ associated to the variable X_{ji} and the coefficients with value $\frac{3}{2}$ associated to the variables X_{mj} and X_{im}). This simple example illustrates that more general and new inequalities for the ATSP problem which are liftings of circuit inequalities but are not obtained by sequential lifting might be suggested by looking closely at combinations of constraints (26) with constraints (23).

The relationships between the LP relaxations of all the models discussed in this paper are summarized in Fig. 2. We have assumed that our conjecture on the equivalence

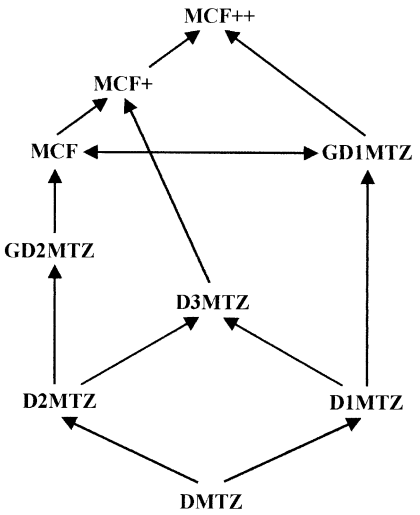


Fig. 2. Relating the LP relaxations of all the models discussed in this paper with respect to the ATSP. An arrow from model A to model B indicates that the LP relaxation of model A is weaker than the LP relaxation of model B. Dominance relationships obtained by transitivity are not shown.

of $GD1MTZ_L$ and $CLIQUE_L$ is true. Thus, we have included an equivalence arrow indicating that the LP relaxations of the models G1DMTZ and MCF are equivalent. It is important to point out that the relations shown below hold only on the space of the X_{ij} variables, i.e, they are valid relationships for the ATSP. However, these relationships do not hold in the space of the X_{ij} and V_{ki} variables and thus, they cannot be used for problems where the V_{ki} variables cannot be removed as the ATSP with precedence costs. The reason for this is that in the space of the X_{ij} and V_{ki} variables, models involving the generalized constraints (26) do not dominate models involving the generalized path inequalities (23) and vice-versa. The results presented in this paper show that a valid relationship between the LP relaxations of the different models in the space of the X_{ij} and V_{ki} variables can be obtained if we drop the equivalence arrow between the models MCF and GD1MTZ and include (17) in the MCF model (otherwise, the model would not be defined on the space of the X_{ij} and V_{ki} variables).

6. Conclusions

In this paper we have proposed generalizations of the lifted DMTZ constraints which have been proposed previously in [10]. One class of these generalized constraints was shown to be derived from the LP relaxation of the MCF model. This derivation has also suggested an augmented MCF model with a tighter LP relaxation which implies a set of lifted circuit inequalities and which is obtained by adding to MCF a disaggregated version of the Desrochers and Laporte inequalities together with a suitable set of linking constraints.

Several open questions have been suggested in the paper. One is related with the alternative transformation which produces a solution feasible for $D1MTZ_L$ from a solution feasible for MCF_L. It is not clear if such a linear transformation exists. However, we have pointed out that different integer linear programming MCF-like models with a strong LP relaxation can be obtained if such a linear transformation exists. Finding the complete characterization of the polytope obtained by projecting $F(MCF_L)$ into the (X_{ij}, V_{ki}) space under the linear transformation (17) and finding the complete characterization of the polytope obtained by projecting $F(MCF+_L)$ into the space of the X_{ij} variables seem also to be two interesting research issues.

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